

API INTEGRATION (CRYPTO PRICE CHECKER)

OBJECTIVE

The objective of this project is to fetch live cryptocurrency price data from the CoinGecko API using Python. The script allows users to search for any cryptocurrency by name and displays the current price in USD. By using the requests module to call the API and parsing the JSON response, this project demonstrates how Python can be used to integrate real-world data into simple, interactive command-line applications.

DESCRIPTIVE

This project is a Python-based cryptocurrency price checker that fetches real-time data using the CoinGecko public API. The script sends a GET request to the API endpoint with parameters for USD currency and retrieves the top 20 coins by market cap. The JSON response is then parsed and filtered based on the user's search input, displaying matching coin names and their current prices. The project demonstrates practical use of the requests library, JSON parsing, and list comprehension for search and filter functionality.

CODE SNIPPET

```
import requests

def get_crypto(search):
    url = "https://api.coingecko.com/api/v3/coins/markets"
    params = {
        "vs_currency": "usd",
        "per_page": 20,
        "page": 1
    }
    response = requests.get(url, params=params)
    data = response.json()
    results = [coin for coin in data if search.lower() in coin["name"].lower()]
    if results:
        for coin in results:
            print(f"Name : {coin['name']}")
```

```

        print(f'Price : ${coin['current_price']}")
        print("-----")
    else:
        print("No coin found!")

search = input("Enter coin name to search: ")
get_crypto(search)

```

SCREENSHOTS

```

PS C:\Users\kanja> pip install requests
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: requests in c:\users\kanja\appdata\roaming\python\python310\site-packages (2.32.5)
Requirement already satisfied: urllib3<3, >=1.22.1 in c:\users\kanja\appdata\roaming\python\python310\site-packages (from requests) (2.6.1)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\kanja\appdata\roaming\python\python310\site-packages (from requests) (2026.2.25)
Requirement already satisfied: idna<4, >=2.5 in c:\users\kanja\appdata\roaming\python\python310\site-packages (from requests) (3.11)
Requirement already satisfied: charset-normalizer<4, >=2 in c:\users\kanja\appdata\roaming\python\python310\site-packages (from requests) (3.4.6)
WARNING: You are using pip version 21.2.3; however, version 26.1.2 is available.
You should consider upgrading via the 'C:\Program Files\Python310\python.exe -m pip install --upgrade pip' command.

PS C:\Users\kanja> python crypto.py
C:\Program Files\Python310\python.exe: can't open file 'C:\Users\kanja\crypto.py': [Errno 2] No such file or directory

PS C:\Users\kanja> cd Desktop
cd : Cannot find path 'C:\Users\kanja\Desktop' because it does not exist.
At line:1 char:1
+ cd Desktop
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (C:\Users\kanja\Desktop:String) [Set-Location],
+ ItemNotFoundException : PathNotFound,Microsoft.PowerShell.Commands.SetLocationCommand
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.SetLocationCommand

PS C:\Users\kanja> ls

Directory: C:\Users\kanja

Mode                LastWriteTime         Length Name
----                -
d-----          13-02-2025         21:41         .anaconda
d-----          20-02-2025         12:28         .conda
d-----          22-12-2025         15:08         .config
d-----          20-02-2025         09:28         .continuum
d-----          27-10-2025         14:00         .eclipse
d-----          24-03-2026         14:18         .idlerc
d-----          27-10-2025         14:58         .p2
d-----          11-04-2026         01:25         .railway
d-----          24-03-2026         14:23         .streamlit
d-----          19-01-2026         11:41         .vscode
d-----          01-06-2026         16:29         .vscode-shared
d-----          13-02-2025         21:41         anaconda
d-----          14-02-2025         14:22         Contacts
d-----          27-09-2025         15:23         CrossDevice
d-----          06-04-2026         06:45         Documents
d-----          02-06-2026         18:41         Downloads
d-----          14-02-2025         14:22         Favorites
d-----          14-02-2025         14:22         Links

```

```

PS C:\Users\kanja> ls

Directory: C:\Users\kanja

Mode                LastWriteTime         Length Name
----                -
d-----          19-01-2026         11:41         .vscode
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d-----          13-02-2025         21:41         anaconda
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d-----          06-04-2026         06:45         Documents
d-----          02-06-2026         18:41         Downloads
d-----          14-02-2025         14:22         Favorites
d-----          14-02-2025         14:22         Links
d-----          14-02-2025         14:22         Music
d-----          09-04-2026         22:58         node_modules
d-----          02-06-2026         11:25         OneDrive
d-----          06-04-2026         06:57         Pictures
d-----          11-04-2026         18:51         portfolio-frontend
d-----          14-02-2025         14:22         Saved Games
d-----          14-02-2025         14:22         Searches
d-----          02-06-2026         21:06         Videos
d-----          12-01-2026         11:13         6529_114_windows.xml
d-----          13-02-2025         21:42         186_condarc
d-----          10-04-2026         22:21         73_gitconfig
d-----          10-04-2026         19:41         0_node_repl_history
d-----          09-04-2026         22:58         29857_package-lock.json
d-----          09-04-2026         22:58         341_package.json

PS C:\Users\kanja> cd OneDrive\Desktop
PS C:\Users\kanja\OneDrive\Desktop> python crypto.py
Enter coin name to search: bitcoin
Name : Bitcoin
Price : $66974

PS C:\Users\kanja\OneDrive\Desktop> python crypto.py
Enter coin name to search: ethereum
Name : Ethereum
Price : $1865.45

PS C:\Users\kanja\OneDrive\Desktop> dogecoin
dogecoin : the term 'dogecoin' is not recognized as the name of a cmdlet, function, script
file, or operable program. Check the spelling of the name, or if a path was included, verify
that the path is correct and try again.
At line:1 char:1
+ dogecoin
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (dogecoin:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

```

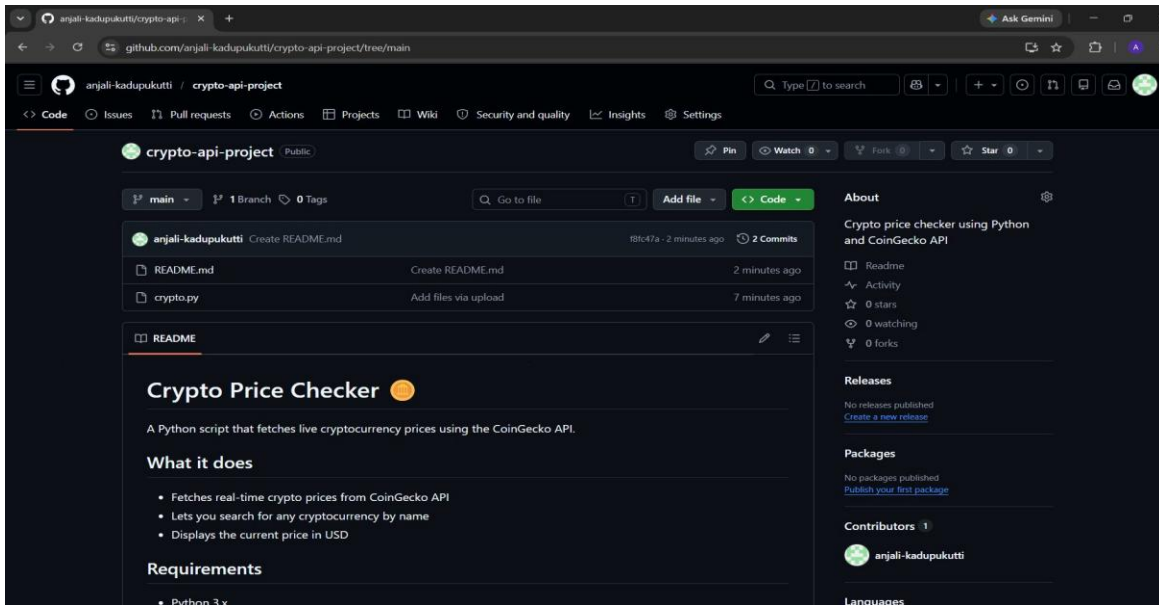
```
File Edit Selection View Go Run Terminal Help
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\kanja\OneDrive\Desktop> dogecoin
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (dogecoin:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

• PS C:\Users\kanja\OneDrive\Desktop> python crypto.py
Enter coin name to search: dogecoin
Name : Dogecoin
Price : $0.093858
-----

• PS C:\Users\kanja\OneDrive\Desktop> python crypto.py
Enter coin name to search: solana
Name : Solana
Price : $74.58
-----

○ PS C:\Users\kanja\OneDrive\Desktop> 
```



GITHUB REPOSITORY LINK

<https://github.com/anjali-kadupukutti/crypto-api-project>

CONCLUSION

This project successfully demonstrates the use of Python for fetching and displaying live cryptocurrency data through API integration. The developed script reduces complexity by providing a simple search interface and retrieving real-time prices from the CoinGecko API without requiring any API key. Through this project, practical experience was gained in Python programming, working with REST APIs, JSON parsing, and version control using Git and GitHub. The solution highlights the power of Python for building lightweight data-driven applications.